

scl-article: The Scientific Computing Laboratory Article Class File

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Abstract

Insert abstract here.

1 Introduction

For quick typesetting instructions please skip to Section 2.

The `scl-article` class was designed for use of the members of the UP Scientific Computing Laboratory.

2 Typesetting

Typesetting with `scl-article` is, well, easy. Just by using the document class entry in the document's preamble as follows: `\documentclass{scl-article}` the typesetting work is nearly done. The `scl-article` class is a relatively conservative extension of the standard `article` class, so most of the environments, section headers, etc. defined by `article` are available.

2.1 Generalities

The following are the general default parameters `scl-article` introduces into the typesetting aspect of articles. Do not alter these – papers deviating from the formatting standards will be automatically rejected.

1. The default paper size is A4 letter. It can be explicitly set to A4 (`a4paper`) or letter (`letterpaper`) paper in the document class entry, e.g., `\documentclass[a4paper]{scl-article}`.
2. The base font size is 11pt (in deference to senior *scientific eyes*¹).
3. The references list is condensed. The default bibliography styles, such as `plain`, `abbrv`, and `alpha`, are suggested.
4. PNG, JPG, and PDF images are supported, i.e., those that are supported by the standard `graphicx` package [2], and render nicely in online versions of PDF documents.

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¹Whatever “scientific eyes” may mean :-)

2.2 Front Matter

The front matter of an `scl-article` article follows the `article` style. For the `\author` command with multiple authors, use `\and` to separate authors from different institutions, as done in this document. If the authors are from the same institution they can be separated by commas or `\\` preceding their institution. If the order of authors from the same institution is not consecutive, follow the same principle as for authors from the separate institutions.

2.3 Section Headings

Section and paragraph headings in `scl-article` are invoked via the standard commands, such as `\section`, `\subsection`, `\subsubsection`, and `\paragraph`. Generally, every non-trivial word must be capitalized according to general capitalization guidelines.

Paragraph headings must have a trailing period. See the examples in this document, e.g., Section 2 is a section, this (Section 2.3) is a subsection, and Section 2.3.1 is a subsubsection.

2.3.1 Subsubsection Header

This is a subsubsection.

Paragraph header. This is a paragraph. One way of saving space when hyper-references are not essential is to use paragraphs instead of subsubsections.

2.4 Page Numbering

Page numbers are at the middle-bottom of every page. Authors must leave the page numbers in as-is.

3 Installation and Usage Instructions

3.1 Installation

The “installation” of the `scl-article` document class is easy. Download the `scl-article.zip` package from the Members’ Resources page of the SCL website and unzip it in the directory where you will prepare your paper.

3.2 Required Packages

The `scl-article` class relies only on packages deemed standard and shipped by most $\text{\LaTeX}$ distributions in the worlds of Linux (`tetex`), MacOS X, and Windows (via Cygwin or MikTeX). If for some reason your distribution is old or doesn’t have the packages listed below, you can always obtain a copy from CTAN [16].

- `inputenc` [9] – with the default option `utf8`, primarily to allow for UTF-8 characters.
- `url` [1] – to provide URL rendering support for the monospaced font, which takes care of special characters as well as line wrapping.
- `hyperref` [13] – to allow hyperlinking of URLs and cross references within an article. Its options are set to either `letterpaper` or `a4paper`, depending on the `\documentclass` options.

- `graphicx` [2] – the standard package for rendering PNG, JPG, and PDF graphic images, primarily in `figure` environments.
- `listings` [11] – to allow highlighted source code listing styles.
- `latexsym` [17] – to provide common math and other symbols.
- `amsthm` [15] – to provide $\mathcal{AMS}$ theorem-like environments.
- `empheq` [8] – to provide equation environments, etc.
- `geometry` [18] – to set `scl-article` margins, outlined in Section 2.1.
- `lastpage` [6] – to allow computationally referencing the last page.
- `fancyhdr` [19] – for running heads.
- `footmisc` [3] – to ensure that footnotes are always at the bottom.

3.3 Recommended Packages

Here is a list of some packages that this guide’s authors have experimented with, and which are suitable for inclusion if needed by article authors. These packages must be loaded using `\usepackage`. In general, authors may use any standard packages provided they do not change the basic layout and font settings established by the `scl-article` class. Such packages must be provided with the submission of articles.

- `rotating` [4] – to rotate floats (figures and tables) on the page, when wide tables or figures do not fit in portrait layout.
- `pdflscape` [12] – similar to `rotating`, but also allows rotating text to make it conveniently viewable in a PDF viewer that supports individual rotated pages. A possible disadvantage is that a page break is forced, which may create gaps before or after the landscape page.
- `algorithm2e` [5] – provides a figure-like algorithm environment for formal algorithm presentation with highlighting.

3.4 Compiling

`pdflatex` [7] is the preferred tool for producing PDF files with `scl-article` class documents.

- Microsoft Windows: TeXnic Center [20] and MikTeX [14] are tools for $\text{\LaTeX}$ processing under Microsoft Windows. The former provides a GUI front-end to $\text{\LaTeX}$, and the latter is the Windows native-compiled `tetex` binaries and standard packages with a comprehensive package update tool.
- Mac OS X: TeXShop [10] is a tool for $\text{\LaTeX}$ processing under Mac OS X. It provides a GUI front-end to $\text{\LaTeX}$.

3.5 Bug Reports

Please report bugs, errors, and omissions you find with the `scl-article` class to its primary author and current maintainer, John Justine Villar, at jsvillar1@up.edu.ph. Any constructive feedback is always welcome.

4 Conclusion

An article that occupies approximately 17 LNCS formatted pages, using the 10pt base font size, takes approximately 14 `scl-article` pages, using the 11pt base font size.

5 Version History

- `scl-article` v. 1.0 – July 2016, initial release

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